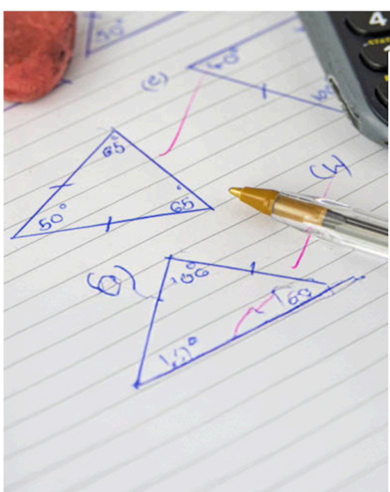
Quantitative Aptitude

Number System

STUDY NOTES

Perfect Square



Perfect Square

A perfect square is defined as an integer that can be expressed as the square of another integer. In other words, a perfect square is a number that results from multiplying an integer by itself.

Conditions to be a Perfect Square:

1. **Last Digit**:
   - If a number ends with **2, 3, 7, or 8**, it cannot be a perfect square.
   - Perfect squares can only have unit digits of **0, 1, 4, 5, 6, or 9**.
2. **Digital Root**:
   - The digital root of a perfect square must be **1, 4, 7, or 9**.
   - If the digital root is **2, 3, 5, 6, 8**, the number is not a perfect square.
3. **Number Ending in 5**:
   - If the unit digit of a perfect square ends in **5**, the tens digit is always **2**.

Finding the Perfect Square Using the Base of 100

This method is useful for quickly calculating the square of numbers close to 100. It relies on adjusting the number based on how far it is from 100, and then applying a simple multiplication.

Steps:

1. **Identify the difference between the number and 100**.
2. **Add or subtract the difference to/from the original number**.
3. **Square the difference.**
4. **Combine the results** by adding the square of the difference to the result of the earlier step.

Example 1: Find the square of 103

1. **Step 1**: Identify the difference from 100.
   Difference = 103 - 100 = 3
2. **Step 2**: Add the difference to the original number.
   103 + 3 = 106
3. **Step 3**: Square the difference.
   $3^2 = 9$
4. **Step 4**: Combine the results.
   $103^2 = 10600 + 9 = 10609$

∴ The correct answer is 10609.

Example 2: Find the square of 97

1. **Step 1**: Identify the difference from 100.
   Difference = 100 - 97 = 3
2. **Step 2**: Subtract the difference from the original number.
   97 - 3 = 94
3. **Step 3**: Square the difference.
   $3^2 = 9$
4. **Step 4**: Combine the results.
   $97^2 = 9400 + 9 = 9409$

∴ The correct answer is 9409.

Example 3: Find the square of 112

1. **Step 1**: Identify the difference from 100.
   Difference = 112 - 100 = 12
2. **Step 2**: Add the difference to the original number.
   112 + 12 = 124
3. **Step 3**: Square the difference.
   $12^2 = 144$
4. **Step 4**: Combine the results.
   $112^2 = 12400 + 144 = 12544$

∴ The correct answer is 12544.

Conclusion:

Using the base of 100 to find the square of numbers close to 100 simplifies the calculation. This method is quick and effective, as it breaks down the square into simpler steps that involve basic addition and squaring small numbers.